

- 9 (a) State what is meant by the de Broglie wavelength.

.....

.....

..... [1]

- (b) Complete Table 9.1 to identify experimental evidence for the wave nature and the particulate nature of matter and of electromagnetic radiation.

Table 9.1

	evidence for	
	wave nature	particle nature
matter		
electromagnetic radiation		

[4]

- (c) An electron in a vacuum is accelerated from rest by a potential difference (p.d.) of 3.5 kV.

For the accelerated electron, calculate

- (i) the speed

speed = ms^{-1} [3]

(ii) the de Broglie wavelength.

wavelength = m [2]

[Total: 10]